

Amar Telidji University Laghouat

Department: Computer Science

Level: 4th Year Engineer

Course: Machine learning, Deep learning and Security

Assignment CH02 : Malware Classification Report

Dataset: Malware Analysis Dataset - API Call Sequences (Kaggle)

- **Supervised by:** Dr. Sarra Boudouh
- **Prepared by:** Smail MERSAD

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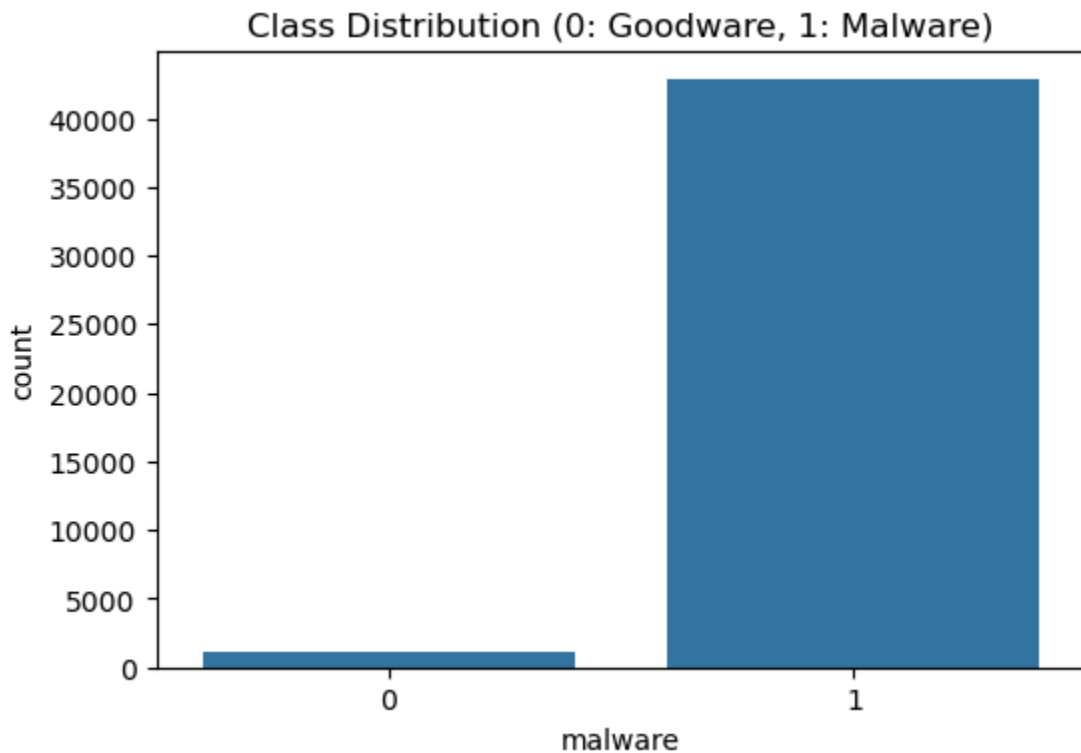
1. Introduction and Methodology

1.1 Objective

The primary goal of this assignment was to apply and evaluate two distinct machine learning algorithms, K-Nearest Neighbors (KNN) and Support Vector Machines (SVM), for the classification of software as either Malware (Class 1) or Goodware (Class 0), based on sequential API call data.

1.2 Data Preprocessing Summary

1. **Exploration:** The dataset contained 43,876 samples and 100 API call features. A critical finding was the severe class imbalance: 97.5% of samples were Malware, and only 2.5% were Goodware.



2. **Feature Engineering:** The non-predictive hash identifier column was removed.

3. **Splitting:** The data was split into 80% for training and 20% for testing.
4. **Normalization:** Feature scaling using StandardScaler was applied to both sets. This step was essential for the distance-based nature of KNN and SVM to prevent features with larger numeric ranges from dominating the models.

1.3 Classification Models

- **K-Nearest Neighbors (KNN):** Trained with (k=5).
- **Support Vector Machine (SVM):** Trained using the Radial Basis Function (RBF) kernel, which is highly effective for non-linear classification boundaries.

2. Analysis of Results

2.1 Overall Performance Comparison

Model	Accuracy	Precision (Malware)	Recall (Malware)
KNN	98.48%	98.55%	99.92%
SVM	98.62%	98.62%	99.99%

The SVM model slightly edged out the KNN model in overall accuracy, precision, and recall for the malware class.

2.2 Deep Dive into Evaluation Metrics

--- KNN Results ---

Accuracy: 0.9848

Precision: 0.9855

Recall: 0.9992

Classification Report:

	precision	recall	f1-score	support
0	0.93	0.44	0.60	226
1	0.99	1.00	0.99	8550
accuracy			0.98	8776
macro avg	0.96	0.72	0.80	8776
weighted avg	0.98	0.98	0.98	8776

--- SVM Results ---

Accuracy: 0.9862

Precision: 0.9862

Recall: 0.9999

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.47	0.64	226
1	0.99	1.00	0.99	8550
...				
accuracy			0.99	8776
macro avg	0.99	0.73	0.81	8776
weighted avg	0.99	0.99	0.98	8776

High Accuracy and Class Imbalance

Both models achieved exceptionally high overall accuracy (above 98.48%). This score is largely misleading due to the severe class imbalance. Given that 97.5% of the test samples are Malware, high accuracy is trivially achieved. Therefore, the focus must shift to the performance metrics of the minority class (Goodware) to properly assess the models' effectiveness.

Malware Detection (Class 1)

The models show robust performance in identifying malicious software:

- **Recall (Malware):** Both models achieved near-perfect Recall (approximately 99.9%). This is highly desirable in malware detection, as it means the system will rarely miss a truly malicious file (low False Negatives).
- **Precision (Malware):** Precision was also very high (approximately 98.6%). This means that when the model flags a file as malware, it is correct nearly 99% of the time.

Goodware Detection (Class 0 - The Critical Issue)

The true vulnerability of the models is revealed in their performance on the minority class (Goodware), which is essential for minimizing disruption to users by flagging legitimate software.

Model	Precision (Goodware)	Recall (Goodware)	F1-Score (Goodware)
KNN	0.93	0.44	0.60
SVM	0.99	0.47	0.64

- **Low Recall:** Both models displayed a significantly low Recall for Goodware (approximately 44% to 47%).
- **Interpretation:** A Recall of 47% means that over half (approximately 53%) of the truly safe files (Goodware) were incorrectly classified as Malware. These are known as **False Positives**.
- **High Precision:** Interestingly, both models had high Precision for Goodware (approximately 93% to 99%). This indicates that when the model did predict a file was Goodware, it was almost always correct. However, the model rarely chose to predict Goodware, resulting in many missed opportunities for correct classification.

2.3 Model Choice and Summary

Model	Strengths	Weaknesses
KNN	High Malware Recall (99.92%), Simplicity of implementation.	Significantly lower Goodware Recall (44%), Computationally expensive during prediction.
SVM	Highest overall metrics (Accuracy 98.62%), Highest Goodware Recall (47%), Excellent Malware Recall (99.99%).	Slower training time.

Conclusion: The **SVM model** delivered the strongest performance across all metrics, marginally improving the detection of the minority Goodware class (47% Recall vs. 44% for KNN).

3. Discussion and Recommendations

3.1 Observed Trade-Off

The current models exhibit a bias toward the majority class (Malware). While this ensures maximum security (almost no threats are missed), it results in a high number of **False Positives** (flagging safe files as malware). In a security context, this trade-off is often acceptable, but a 53% False Positive rate for legitimate files would be highly disruptive in a production environment.

3.2 Future Recommendations

To build a more balanced and reliable malware classification system, future work should focus on addressing the class imbalance issue:

1. **Sampling Techniques:** Employ techniques like **SMOTE (Synthetic Minority Over-sampling Technique)** to generate synthetic samples for the Goodware class, thus balancing the training data.
2. **Algorithm Tuning:** Experiment with **class weights** within the SVM model. By assigning a higher penalty (weight) to misclassifications of the minority (Goodware) class, the model can be explicitly forced to pay more attention to correctly identifying safe files.
3. **Alternative Metrics:** Continue to prioritize **Recall for Malware** (to minimize threats) and **Recall for Goodware** (to minimize False Positives) over general accuracy.